WORKSHEET

Calculating quantities in ratios

1 Express each ratio in its simplest form.

a 4 : 10

b 10 : 20

c 10 : 5

d 12 : 14

e 24 : 20

f 100 : 300

g 200 : 500

h 12 : 9

i 8 : 100

j 9 : 6

2 Write each quantity in the same units and then simplify the ratio.

a 20 minutes : 1 hour

b $3.00 : 60c

c 4 days : 2 weeks

d 2 metres : 150 cm

3 Complete the missing values.

a 1 : 3 = 5 : _____

b 1 : 8 = 3 : _____

c 1 : 20 = 7 : _____

d 1 : 9 = 4 : _____

e 2 : 3 = 8 : _____

f 2 : 5 = 12 : _____

g 2 : 9 = 20 : _____

h 50 : 3 = _____ : 12

i 5 : 2 = _____ : 20

j 9 : 5 = _____ : 15

4 Ian is mixing orange concentrate with water in the ratio of 1 : 4.

a How much water will he mix with 80 mL of concentrate?

b How much concentrate should he add to 400 mL water?

c Ian needs to make 1 litre of mixture. How much concentrate and water should he mix?

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5 Grant is mixing water and poison in the ratio of 40 : 1 to make a weed spray.

a How much water should he mix with 80 mL of poison?

b How much poison should he add to 600 mL of water?

c How much water and poison should Grant mix to make 2 L of weed spray?

6 Caroline is blending wool and silk in the ratio of 5 : 2 to make the yarn to knit a scarf. She has 50 g of silk. How much wool will she need?

7 Peter's lawn mower saw uses petrol and oil in the ratio of 50 : 1. How much oil should Peter add to each litre of petrol he mixes to use in his lawn mower?

8 The fuel for Jim's chainsaw is a 25 : 1 petrol : oil mixture. How much oil and petrol is required to make 1.3 L of fuel?

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Answers

1 **a** 2 : 5
b 1 : 2
c 2 : 1
d 6 : 7
e 6 : 5
f 1 : 3
g 2 : 5
h 4 : 3
i 2 : 25
j 3 : 2

2 **a** 1 : 3
b 5 : 1
c 2 : 7
d 4 : 3

3 **a** 15
b 24
c 140
d 36
e 12
f 30
g 90
h 200
i 50
j 27

4 **a** 320 mL
b 100 mL
c 200 mL concentrate, 800 mL water

5 **a** 3200 mL
b 15 mL
c 48.8 mL poison, 1951.2 mL water

6 125g

7 20 mL

8 1250 mL petrol, 50 mL oil

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